Year 8 Science – Booklet 2: Physics

Magnetism & Electromagnetism — ANSWERS

Quick Test (p.119)

1. They repel each other (push apart).
2. In a magnetised piece of steel, all the mini-magnets in every domain point in the same direction (they are all aligned); in an unmagnetised piece, different domains point in different, random directions.
3. Increase the current flowing through the coil, and increase the number of turns of wire on the coil (also accept: add a soft iron core).
4. Any two of: an electric bell, a scrapyard crane (for lifting/sorting cars and scrap metal), an electric motor, a relay switch, an MRI scanner, a loudspeaker.

Section A – Multiple Choice (5 marks)

1. c) steel
2. a) copper
3. b) that the field is strong here
4. b) it can be turned on and off
5. c) using thicker wire

Section B (3 marks)

1. Complete the diagrams of molecular magnets (2 marks)

a). Unmagnetised: the domains (the wavy-line-divided regions) each contain small arrows pointing in different, mixed directions – no overall alignment.

b). Magnetised: all the small arrows in every domain point the same way (fully aligned in one direction).

2. Complete the magnetic field diagram around a bar magnet (1 mark)

Field lines should emerge from the N pole, curve around the outside of the magnet, and enter the S pole – densest (closest together) near the poles, spreading further apart away from the magnet.

Section C (10 marks)

1. Simple electromagnet

a). The field lines should form the same pattern as a bar magnet: emerging in loops from one end of the coil, curving round the outside, and entering the other end.

b). Increase the current flowing through the coil; increase the number of turns of wire on the coil (also accept: add a soft iron core).

c). It can be turned on and off (by switching the current on/off), unlike a permanent magnet.

2. Explain what happens when the bell-push is pressed (4 marks)

Pressing the bell-push completes the circuit, so current flows through the electromagnet, magnetising it. The magnetised electromagnet attracts the soft iron armature towards it, and as it moves, the attached hammer strikes the gong, making a sound. This movement also breaks the contact at C, opening the circuit and turning the electromagnet off. With no magnetism, the springy metal strip pulls the armature back to its original position, which remakes the contact at C and completes the circuit again. As long as the bell-push stays pressed, this cycle repeats rapidly, so the armature vibrates back and forth and the hammer keeps striking the gong.

Note: This is a proprietary Aristeia Academy Tuition worksheet, not an exam-board past paper, so no publicly published markscheme exists for it. These answers were composed directly from the teaching content on the preceding pages of the booklet.